Free balancing equations worksheet worksheet 24

Maths | 10th grade | Algebra

1. If 3x - 4y = 12 and x + y = 5, find the value of x.

A) 2
B) 3
C) 4
D) 5

2. Simplify the expression: (2x² - 3x + 4) - (x² + 2x - 5).

A) x² - 5x + 9
B) x² - 5x + 1
C) 3x² - 5x + 9
D) x² + x + 9

3. Factor the quadratic equation: x² - 5x + 6.

A) (x - 2)(x - 3)
B) (x + 2)(x + 3)
C) (x - 1)(x - 6)
D) (x + 1)(x - 6)

4. Solve for x: 4^(2x - 1) = 16.

A) 0.5
B) 1
C) 1.5
D) 2

5. Find the vertex of the parabola given by the equation y = x² - 4x + 3.

A) (2, -1)
B) (2, 3)
C) (1, -4)
D) (3, -2)

6. If f(x) = 2x - 1 and g(x) = x², find (f " g)(2).

A) 3
B) 5
C) 7
D) 9

7. What is the sum of the roots of the equation 3x² - 12x + 9 = 0?

A) 3
B) 4
C) 5
D) 6

8. Solve the inequality: 2/(x - 3) "e 1.

A) x "d 2 or x > 3
B) x > 3

C) x < 2 or x > 3
D) x "e 3

9. If 4^x = 8^(x-1), find the value of x.

A) 0
B) 1
C) 2
D) 3

10. If the difference between the squares of two consecutive integers is 9, find the integers.

A) 4 and 5
B) 5 and 6
C) 6 and 7
D) 7 and 8

Answer Key

1. B
Solve the system of equations to find x. Substitute x = 5 - y into the first equation.

2. B
Distribute the negative sign and combine like terms.

3. A
Find two numbers that multiply to 6 and add to -5.

4. B
Rewrite 16 as a power of 4 and equate the exponents.

5. A
Use the vertex formula x = -b/2a to find the x-value, then substitute back to find y.

6. B
First find g(2), then substitute that result into f(x).

7. D
The sum of the roots is given by -b/a.

8. A
Solve the inequality and consider the domain where the expression is defined.

9. C
Express both sides with a common base and equate the exponents.

10. C
Let the integers be n and n+1. Solve $(n+1)^2 - n^2 = 9$.